9 **(a)** A uranium-235 nucleus has a radius of 8.9×10^{-15} m.

Calculate, for a uranium-235 nucleus,

(i) its mass,

mass = ... kg [1]

(ii) its mean density.

density = ... kg m^{-3} [2]

(iii) The density of a lump of uranium is 1.9×10^{4} kg m^{-3}.

Using your answer to **(a)(ii)**, state and explain what can be inferred about the structure of the atom.

.....

.....

.....[2]

(b) In a nuclear reaction, a possible fission reaction is

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{144}_{56}Ba + {}^{90}_{36}Kr + 2\,{}^{1}_{0}n.$$

(i) Explain what is meant by a fission reaction.

.....

.....

.....[2]

(ii) The binding energy per nucleon of each nucleus is as follows:

$^{235}_{92}U$: 1.2191×10^{-12} J

$^{144}_{56}Ba$: 1.3341×10^{-12} J

$^{90}_{36}Kr$: 1.3864×10^{-12} J

Use these data to calculate

1. the energy released in this fission reaction (give your answer to three significant figures),

energy = ... J [3]

2. the mass equivalent of this energy.

mass = ... kg [2]

(iii) Suggest why the neutrons were not included in your calculations in **(ii) 1.**

.....

.....[1]

(c) Strontium-90 is a radioactive isotope having a half-life of 28.0 years. Strontium-90 has a density of 2.54 g cm^{-3}.

A sample of Strontium-90 has an activity of 6.4×10^9 Bq.

(i) Define decay constant.

.....

.....[1]

(ii) Calculate

1. the decay constant λ of Strontium-90,

λ = ... s^{-1} [1]

2. the mass of Strontium-90 in the sample,

mass = ... g [3]

3. the volume of Strontium-90 in the sample.

volume = ... cm^3 [1]

(iii) By reference to your answer in **(c)(ii)3.**, suggest why dust that has been contaminated with Strontium-90 presents a serious health hazard.

.....

.....

.....[1]

[Total: 20]